

Overview of the course

- Course Name: Introduction to Computational Chemistry
- Course Code: CHY1005
- LTP Course: Lecture, Tutorial, and practical
- 4 Credits; 3 Sessions per week

AB226

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Overview of the course

Sl.No.	Mark Title	Max. Mark	Weightage %
1	Mid Term	50	30
2	Class Assesment - Tutorial	5	5
3	Class Assesment - Group Activity	5	5
4	Lab Assessment - Continuous Assessment	10	10
5	Lab Assessment - Challenging Task	10	10
6	Lab Assessment - Viva	5	5
7	Attendance	5	5
8	Term End Examination	100	30

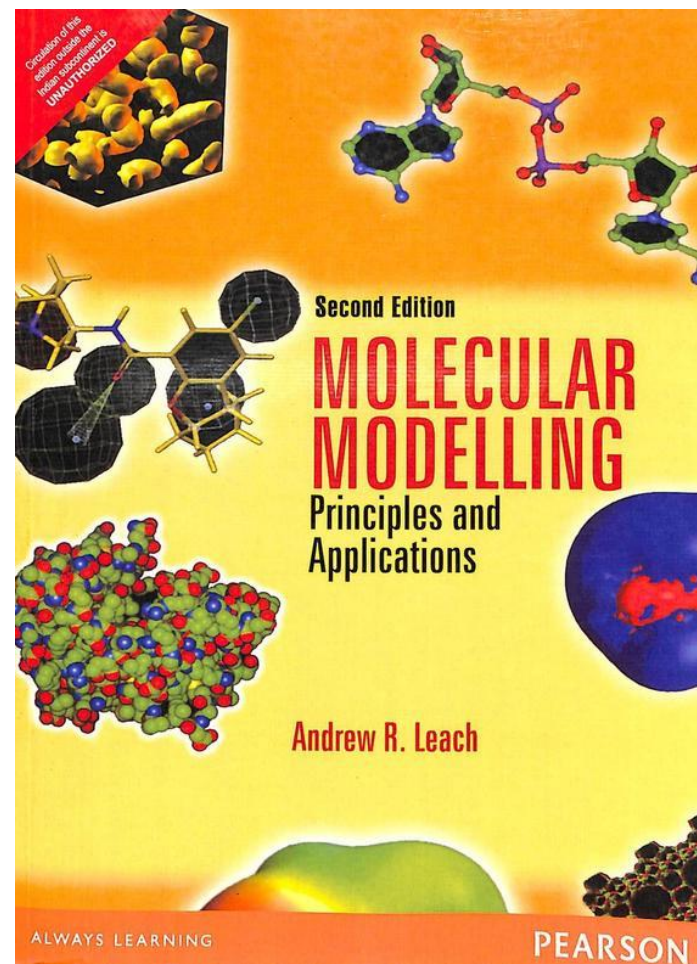
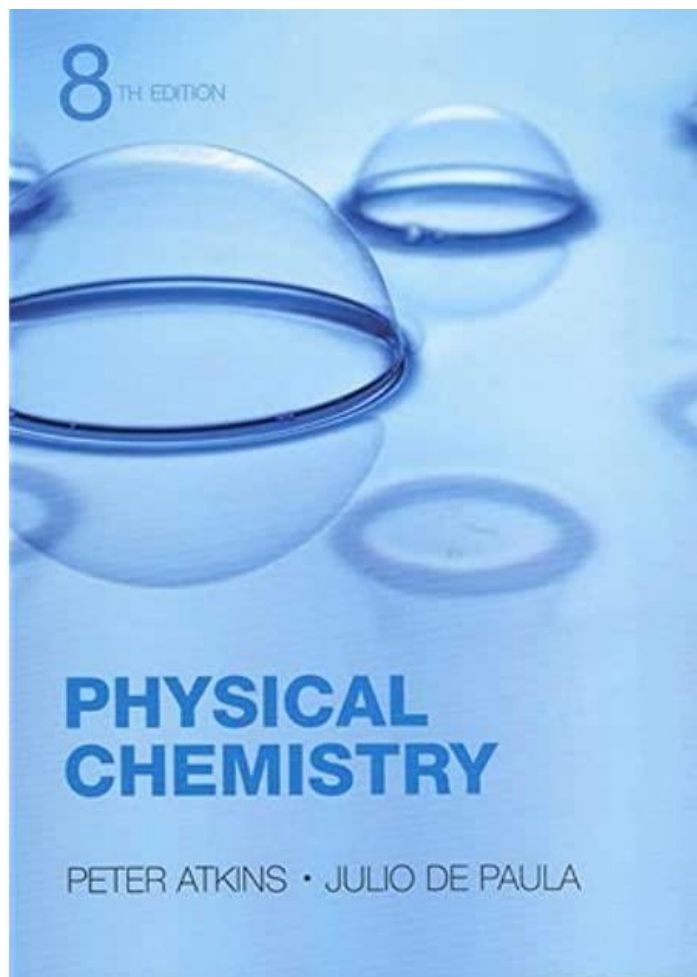
Course structure

- **Module 1: Introduction:** Overview of the course, history and promises of computational chemistry, tools for computational chemistry, units, errors in computed quantities.
- **Module 2: Quantum Chemistry:** Historical development, Bohr's atomic model, de Broglie wavelength, Heisenberg uncertainty principle, Schrodinger equation, Wave function, particle in a box, Hydrogen atom, radial and angular solution to hydrogen atom, applications and limitations.
- **Module 3: Thermodynamics:** Intensive and extensive variables, state and path functions, Laws of Thermodynamics (First law and enthalpy; second law and entropy, spontaneity, and equilibrium; third law and absolute entropy) free energy, Gibbs and Maxwell's relations, Ideal and real gases.

Course structure

- **Module 4: Potential Energy Surfaces:** Chemical bonds and intermolecular interactions, Types of intermolecular interactions (charge distribution of isolated molecules, electrostatic interaction, induction interaction, London or dispersion forces, hydrogen bonding, repulsive interaction, relative contribution of different terms), representing the potential energy surfaces (pair additivity, rare gas), intramolecular interactions (bond stretching, angle bending, torsional and improper terms).
- **Module 5: Molecular Dynamics:** Introduction to ensembles, force fields, integration of Newton's laws of motion, force calculation, energy minimization, periodic boundary conditions, choice of input configuration, velocities, and time-step, applications, and calculation of simple thermodynamic variables.
- **Guest Lectures**

Books



Introduction: Measurements

Measurements

- Chemistry is about observation! Therefore, all lab experiments involve data collection.
- There are two types of data we can collect in any lab: qualitative and quantitative.
- Qualitative:
 - Descriptive, non-numerical data
 - (ex) color, phase, texture
- Quantitative:
 - Numbers with units



Units of measurement

The standard used for the measurement of a physical quantity is called a unit.

Examples:

- metre, foot, inch for length
- kilogram, pound for mass
- second, minute, hour for time
- fahrenheit, kelvin for temperature

Physical Quantity	Name of Unit	Abbreviation
Mass	Kilogram	kg
Length	Meter	m
Time	Second	s ^a
Temperature	Kelvin	K
Amount of substance	Mole	mol
Electric current	Ampere	A
Luminous intensity	Candela	cd

^aThe abbreviation sec is frequently used.

- Numerous units of measurement based on different standards.
- Need for a well-defined system based in precises, reproducible standards.

Dimensions of a physical quantity

The powers of fundamental quantities in a derived quantity are called dimensions of that quantity.

To check the correctness of equation

To convert units

To derive a formula

Fundamental quantities

Dimensions

Length

[L]

Mass

[M]

Time

[T]

Temperature

[K]

Current

[A]

Amount of substance

[N]

Luminous intensity

[J]

Example:

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

$$= \frac{\text{Mass}}{\text{length} \times \text{breadth} \times \text{height}}$$

$$[\text{Density}] = \frac{[M]}{[L] \times [L] \times [L]} = \frac{[M]}{[L^3]} = [ML^{-3}]$$

Hence the dimensions of density are 1 in mass and -3 in length.

Prefix-Based Unit System

A larger or smaller unit is indicated by a SI prefix –

a **prefix** used in the International System to indicate a **power of 10**.

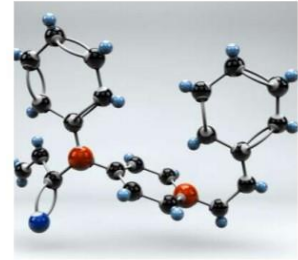
Factor	Name	Symbol	Factor	Name	Symbol
10^{24}	yotta	Y	10^{-1}	deci	d
10^{21}	zetta	Z	10^{-2}	centi	c
10^{18}	exa	E	10^{-3}	milli	m
10^{15}	peta	P	10^{-6}	micro	μ
10^{12}	tera	T	10^{-9}	nano	n
10^9	giga	G	10^{-12}	pico	p
10^6	mega	M	10^{-15}	femto	f
10^3	kilo	k	10^{-18}	atto	a
10^2	hecto	h	10^{-21}	zepto	z
10^1	deka	da	10^{-24}	yocto	y

1 micron = 10^{-6} m



Bacterias

1 nanometer = 10^{-9} m



Molecules

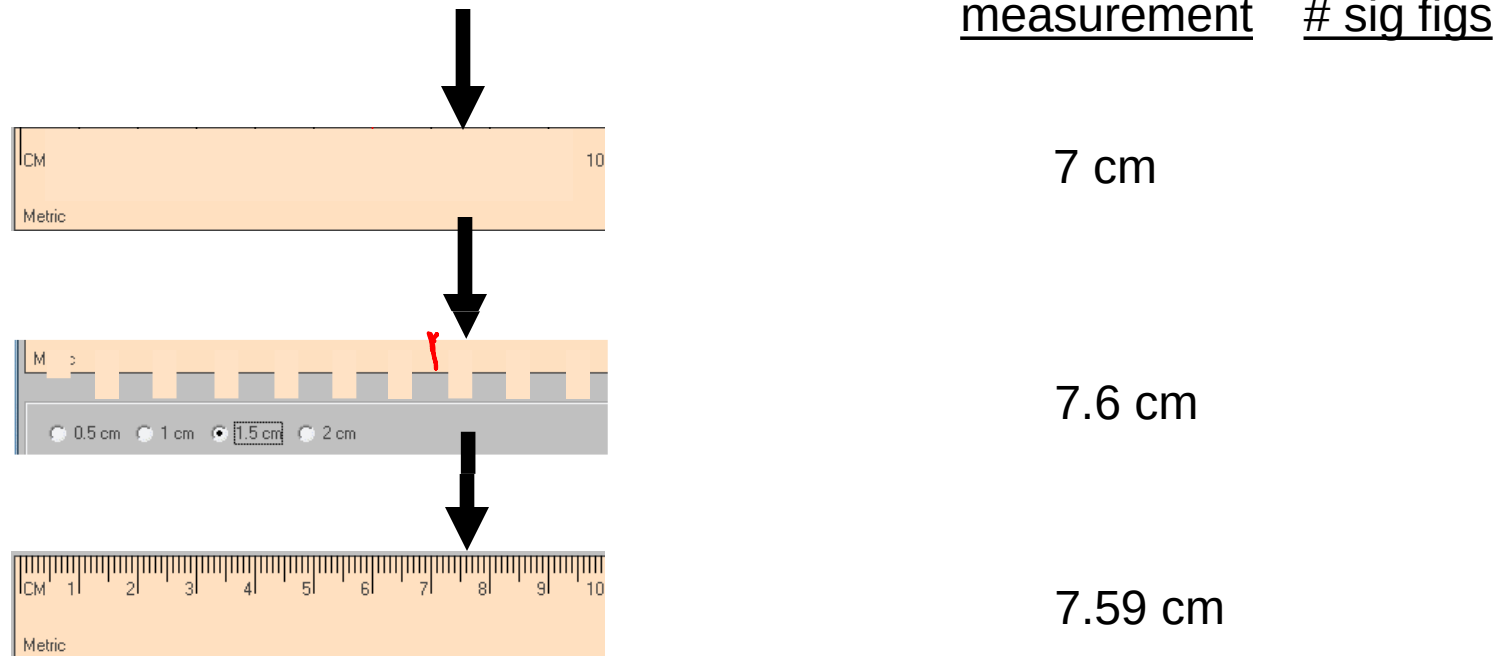
- Astronomical unit = It is defined as the mean distance of the earth from the sun.
- 1 astronomical unit = 1.5×10^{11} m



Distance of planets

Significant Figures

- **Significant figures** are those digits in a measured number (or result of the calculation with a measured number) that include all certain digits plus a final one having some uncertainty (first digit basically guessing).
- The significant digit farthest to the right in the measured quantity is the uncertain one.



Rules for counting significant figures

1

All non-zero digits are significant.

Number	Significant figures
16	2
35.6	3
6438	4

2

Zeros between non-zero digits are significant.

Number	Significant figures
<u>2</u> 05	3
3 <u>00</u> 8	4
6 <u>0.00</u> 5	5

3

Terminal zeros in a number without decimal are not significant unless specified by a least count.

Number	Significant figures
<u>400</u>	1
30 <u>50</u>	3
(20 ± 1) s	2

4

Terminal zeros that are also to the right of a decimal point in a number are significant.

Number	Significant figures
64. <u>00</u>	4
3.6 <u>0</u>	3
25.06 <u>0</u>	5

5

If the number is less than 1, all zeroes before the first non-zero digit are not significant.

Number	Significant figures
<u>0.00</u> 64	2
<u>0.0</u> 850	3
<u>0.000</u> 2050	4

6

During conversion of units use powers of 10 to avoid confusion.

Number	Significant figures
2.700 m	4
2.700 × 10 ² cm	4
2.700 × 10 ⁻³ km	4

Exact quantities

- In certain cases, some situations will utilize relationships that are exact, defined quantities.
 - For example, a dozen is defined as exactly 12 objects (eggs, cars, donuts, whatever...)
 - 1 km is defined as exactly 1000 m.
 - 1 minute is defined as exactly 60 seconds.
- Each of these relationships involves an infinite number of significant figures following the decimal place when being used in a calculation.

Relationships between metric units are **exact** (e.g. 1 m = 1000 mm, exactly)

Relationships between imperial units are **exact** (e.g. 1 yd = 3 ft, exactly)

Relationships between metric and imperial units are **not exact** (e.g. 1.00 in = 2.54 cm)

Rules: Significant Figures

Addition & subtraction

The final result would round to the same decimal place as the least precise number.

Example:

- $13.2 + 34.654 + 59.53 = 107.384 = 107.4$
- $19 - 1.567 - 14.6 = 2.833 = 3$

Multiplication & division

The final result would round to the same number of significant digits as the least accurate number.

Example:

- $1.5 \times 3.67 \times 2.986 = 16.4379 = 16$
- $6.579 / 4.56 = 1.508 = 1.51$

Rules for rounding off a measurement

1

If the digit to be dropped is less than 5, then the preceding digit is left unchanged.

Number	Round off up to 3 digits
64.62	64.6
3.651	3.65
546.3	546

2

If the digit to be dropped is more than 5, then the preceding digit is raised by one.

Number	Round off up to 3 digits
3.479	3.48
93.46	93.5
683.7	684

3

If the digit to be dropped is 5 followed by digits other than zero, then the preceding digit is raised by one.

Number	Round off up to 3 digits
62.354	62.4
9.6552	9.66
589.51	590

4

If the digit to be dropped is 5 followed by zero or nothing, the last remaining digit is increased by 1 if it is odd, but left as it is if even.

Number	Round off up to 3 digits
53.350	53.4
9.455	9.46
782.5	782

Example

$$\frac{[104.6 \times (68.2 + 14)]}{22.58} =$$

$$\frac{[104.6 \times 82.2]}{22.58} =$$

$$\frac{8598.12}{22.58} = 380.7847652790079716563330380868 \dots$$

$= 380$ or 3.8×10^2  Our final answer should be reported with 2 sig figs.

Example

- Sample data:

height (h) = 1.58 cm

diameter = 0.92 cm; radius (r) = 0.46 cm

$$\text{Volume} = \pi r^2 h = \pi (0.46 \text{ cm})^2 (1.58 \text{ cm})$$

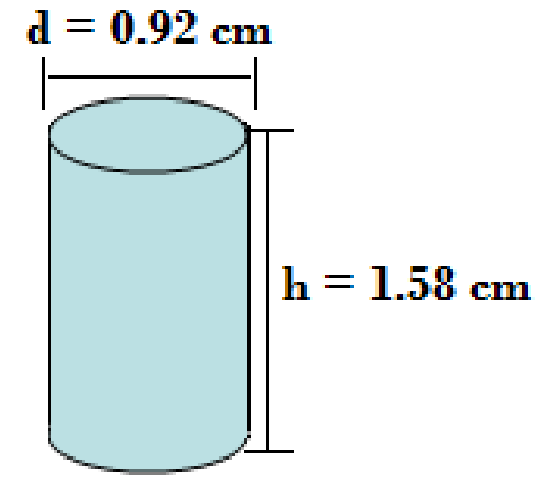
2 sig figs 3 sig figs

$$= 1.050322389 \text{ cm}^3$$

Only operation here
is multiplication

$$\text{Answer} = 1.1 \text{ cm}^3$$

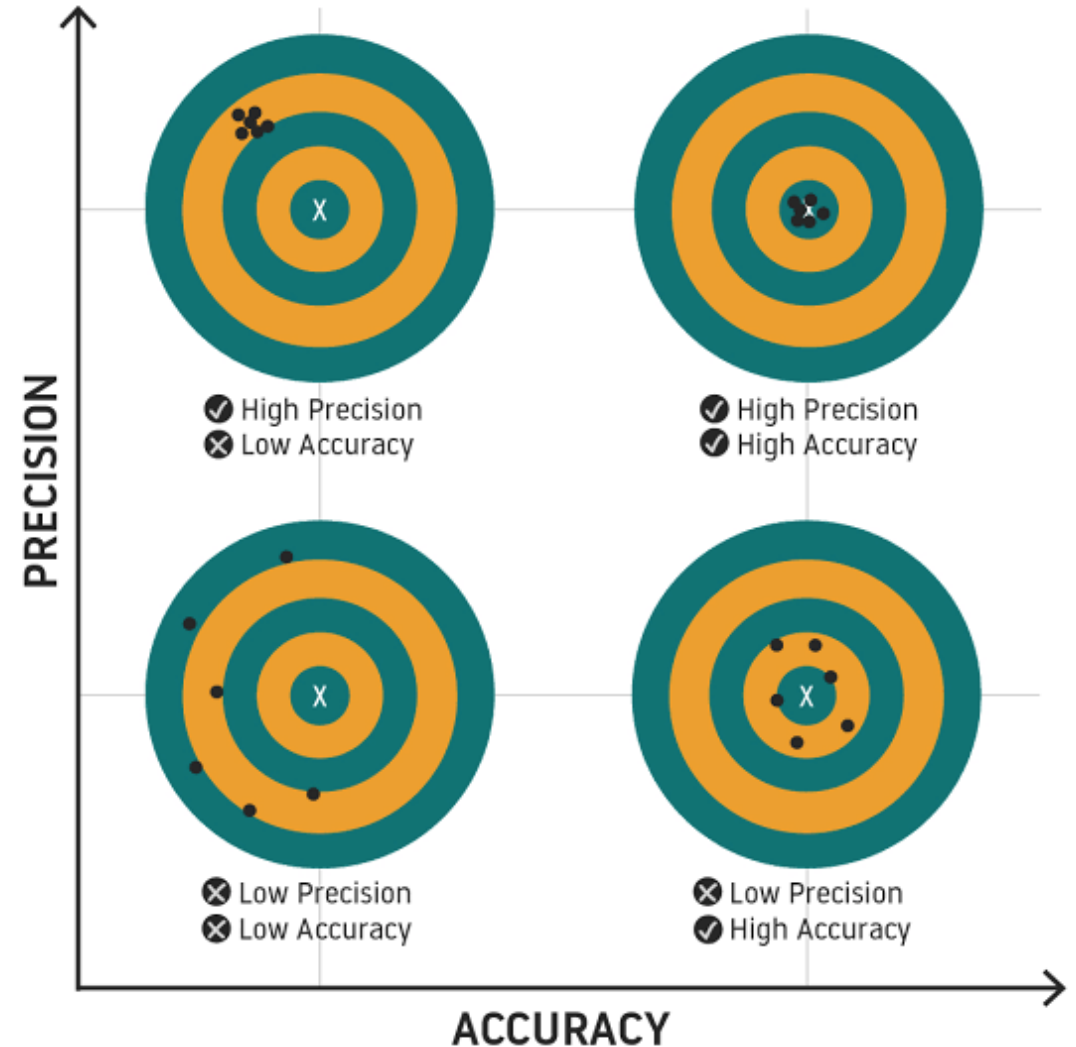
you should round your
answer to 2 sig figs



$$V = \pi r^2 h$$

Good measurement?

- Any good data that is collected needs to be **both precise and accurate**.
- Accuracy:**
 - proximity of your results to the actual answer
- Precision:**
 - getting the same results for multiple trials.



Error

- **Measured values are never absolutely accurate**, they always have error.
- For example, a balance that reads to the nearest 0.001 g is more accurate than one that reads to the nearest 0.01 g.

Why?

Random and systematic errors

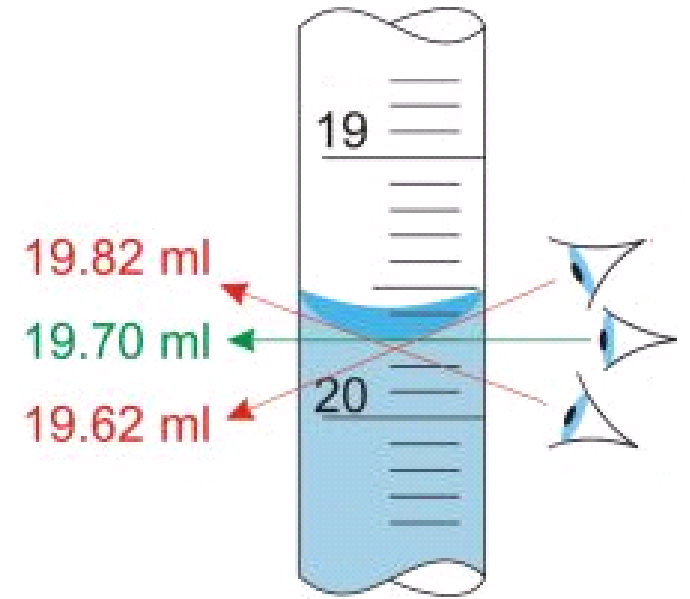


Random and systematic errors

Random: random fluctuations in the measured data due to:

- the readability of the instrument
- the effects of something changing in the surroundings between measurements
- the observer being less than perfect

- Random errors can be reduced by averaging.



Systematic errors is error due to the instrument being “out of adjustment.”

- An instrument with a zero-offset error.
- A meter stick might be worn off or rounded at one end
- An instrument that is improperly calibrated

- Systematic errors can be detected using **different methods** of measurement

Error:

- Value of error?

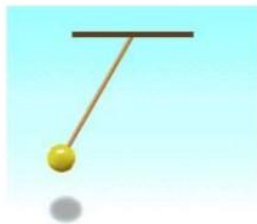
Errors in measurement

Difference between the actual value of a quantity and the value obtained by a measurement is called an error.

$$\text{Error} = |\text{actual value} - \text{measured value}|$$

Error analysis

For example, suppose you measure the oscillation period of a pendulum with a stopwatch five times.



Trial no (i)	1	2	3	4	5
Measured value (X_i)	3.9	3.5	3.6	3.7	3.5

The average of all the five readings gives the most probable value for time period.

$$\bar{X} = \frac{1}{n} \sum X_i$$

$$\bar{X} = \frac{3.9 + 3.5 + 3.6 + 3.7 + 3.5}{5} = \frac{18.2}{5}$$

$$\bar{X} = 3.64 = 3.6$$

The arithmetic mean of all the absolute errors is called mean absolute error.

$$\Delta\bar{X} = \frac{1}{n} \sum \Delta X_i$$

$$\Delta\bar{X} = \frac{0.3 + 0.1 + 0 + 0.1 + 0.1}{5} = \frac{0.6}{5}$$

$$\Delta\bar{X} = 0.12 = 0.1$$

Result = best estimate $\pm$ error

The magnitude of the difference between mean value and each individual value is called absolute error.

$$\Delta X_i = |\bar{X} - X_i|$$

The absolute error in each individual reading:

X_i	3.9	3.5	3.6	3.7	3.5
ΔX_i	0.3	0.1	0	0.1	0.1

The relative error is defined as the ratio of the mean absolute error to the mean value.

$$\text{relative error} = \Delta \bar{X} / \bar{X}$$

$$\Delta \bar{X} / \bar{X} = \frac{0.1}{3.6} = 0.0277$$

$$\Delta \bar{X} / \bar{X} = 0.028$$

The relative error multiplied by 100 is called as percentage error.

$$\text{percentage error} = \text{relative error} \times 100$$

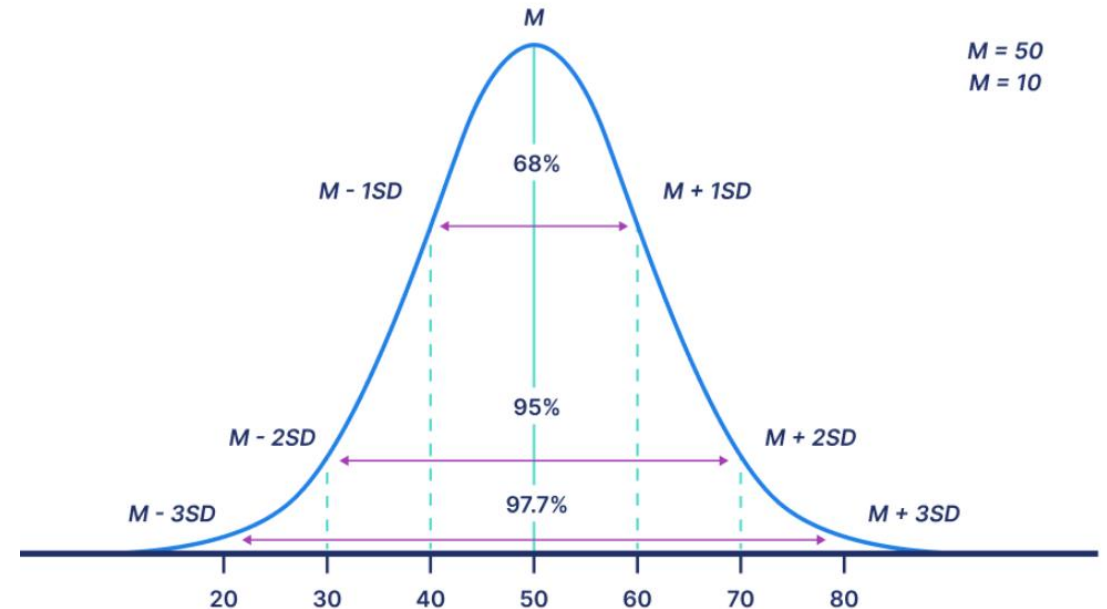
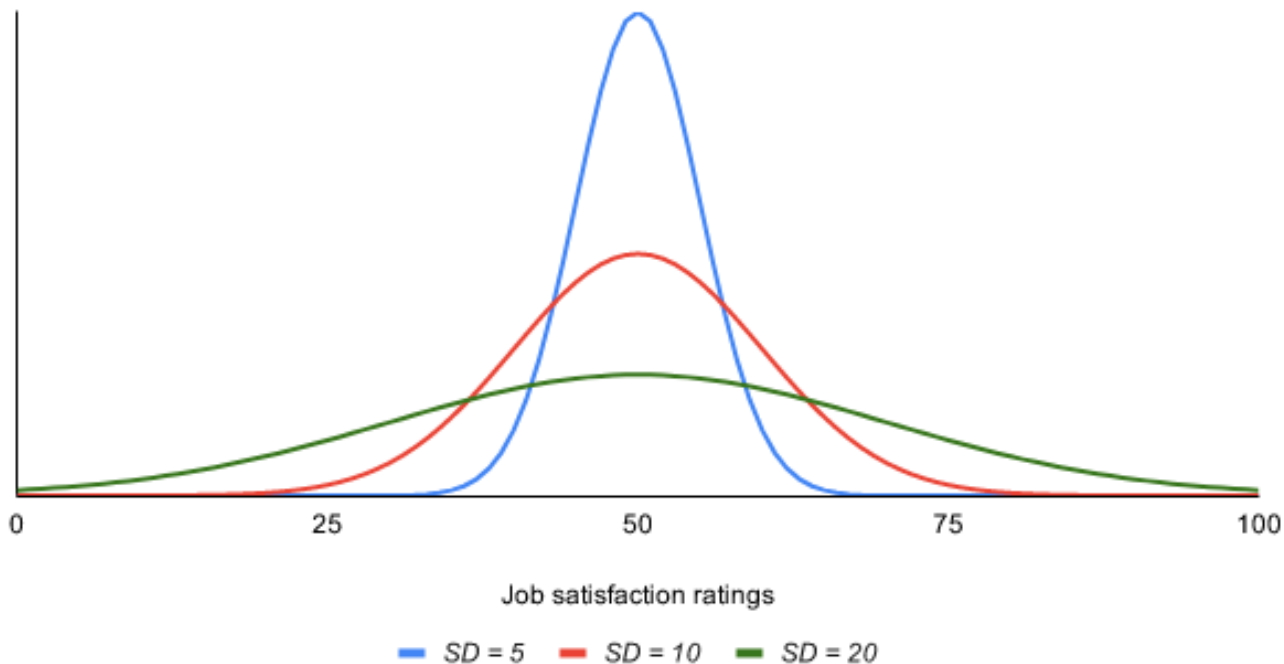
$$\text{percentage error} = 0.028 \times 100$$

$$\text{percentage error} = 2.8 \%$$

Standard Deviation

The **standard deviation** is the average amount of variability in your dataset. It tells you, on average, how far each value lies from the mean.

$$s = \sqrt{\frac{(x_i - \bar{x})^2}{n-1}}$$



Class Interval	Mid-Value (m)	Frequency (f)	fm	$x = m - \bar{X}$ 	x^2	fx^2
0 - 10	5	2	10	-20	400	800
10 - 20	15	7	105	-10	100	700
20 - 30	25	10	250	0	0	0
30 - 40	35	5	175	10	100	500
40 - 50	45	3	135	20	400	1200
		N = 27	Σfm = 675			Σfx² = 3200

$$\text{Mean}(\bar{x}) = \frac{\sum x_i f_i}{\sum f_i}$$

$$\sigma = \sqrt{\frac{\sum fx^2}{N}}$$

$$= \sqrt{\frac{3200}{27}}$$

$$= \sqrt{118.52}$$

$$= 10.89$$

Combination of errors

In different mathematical operations like addition, subtraction, multiplication and division the errors are combined according to some rules.

- Let ΔA be absolute error in measurement of A
- Let ΔB be absolute error in measurement of B
- Let ΔX be absolute error in measurement of X

When $X = A \pm B$

$$\frac{\Delta X}{X} = \frac{\Delta A + \Delta B}{A \pm B}$$

$$\Delta X = \Delta A + \Delta B$$

When $X = A \times B$ or A / B

$$\frac{\Delta X}{X} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$$

$$\Delta X = \left(\frac{\Delta A}{A} + \frac{\Delta B}{B} \right) X$$

An experimentalist makes independent measurements of the length and height of a rectangular feature. The values and their standard deviations are 10.62 ± 0.46 microns and 12.46 ± 0.52 microns. Calculate the perimeter P and the area A of the feature including the standard deviation of each.

$$\begin{aligned} P &= 2 * (10.62 + 12.46) \pm 2 *(0.46 +0.52) \\ &= 46.16 \pm 1.96 \text{ microns} \end{aligned}$$

$$\begin{aligned} A &= 10.62 * 12.46 \pm ((0.46/10.62) + (0.52/12.46)) * 10.62 * 12.46 \\ &= 132.33 \pm 11.11 \text{ square microns} \end{aligned}$$

Suppose you measure three numbers as follows: $x = 200 \pm 2$, $y = 50 \pm 2$, $z = 40 \pm 2$. Use step-by-step propagation to find the quantity $q = x/(y - z)$ with its uncertainty.

$w = (4.52 \pm 0.02)\text{cm}$, $x = (2.0 \pm 0.2)\text{cm}$, $y = (3.0 \pm 0.6)\text{ cm}$. Find $z = x + y - w$ and its uncertainty

It takes 10.5(1) s for a sprinter to run 100.00(5) m. Calculate the average speed of the sprinter in meters per second and km/hr.

A student measures three lengths a , b and c in cm and a time t in seconds:

$$a = 50 \pm 4; \quad b = 20 \pm 3; \quad c = 70 \pm 3; \quad t = 2.1 \pm 0.1$$

Calculate

- $a + b + c$,
- a / t ,
- $(a + c) / t$